

A Relational Observable Substrate and its NCG Realization

The Universal Calculus, the Connes–Chern Character,
and the Disclinated Haldane Class

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Abstract

The relational substrate is a nest of local observable algebras carrying its own descent data: to each patch U_i a local algebra A_i , to each overlap U_{ij} an algebra A_{ij} , with restriction maps $\rho_{ij}^i: A_i \rightarrow A_{ij}$ and a comparison $\ast$ -isomorphism $\alpha_{ij}: A_i|_{U_{ij}} \xrightarrow{\sim} A_j|_{U_{ij}}$ that is strict on triple overlaps, $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$. These data descend directly to a relational algebra-sheaf $\mathcal{A}_{\text{rel}}$ over the comparison site $K = N(R_Q)$; the sheaf is genuinely relational, and flat, before it is spectral. Noncommutative geometry enters at one place: the operator nest $\mathcal{O}p_{\text{rel}} = \text{Op}_{\text{NCG}}(\mathcal{A}_{\text{rel}})$, which represents the local algebras, implements the comparisons by functorial unitaries V_{ij} , and carries a faithful trace T . Occupied subspaces and Fermi projections are selected inside $\mathcal{O}p_{\text{rel}}$, downstream of the substrate. The differential is the universal calculus $\Omega_u^\bullet$, carried for free; curvature is the Grassmann/Bellissard form $F = P(dP)^2$, born only when d acts on the selected projection, with $F_{ijk}^{\text{rel}} = P_i dV_{ij} dV_{jk}$. Because the implementers compose strictly, $V_{ij}V_{jk} = V_{ik}$, the substrate carries no holonomy and the Bianchi identity is automatic. A cyclic characteristic map χ_τ pairs the curvature into a Čech–Connes class $c_1^{\text{rel}} \in \check{H}^2(N(R_Q); \mathbb{Z})$; swapping τ selects Chern, $\mathbb{Z}_2$, gap-labeling, or Floquet invariants on the same substrate. Under a crystalline comparison hypothesis c_1^{rel} recovers the Bloch–Berry Chern class of the Haldane model, and localizes at a pentagonal disclination to a transport-determinant holonomy.

Architecture. Two flat stages, a universal differential, and a characteristic map:

$$\begin{array}{ccc} \underbrace{\mathcal{N}_{\text{rel}} = (\{A_i\}, \{A_{ij}\}, \rho_{ij}, \alpha_{ij}), \quad \alpha_{ij}\alpha_{jk} = \alpha_{ik}}_{\text{Stage 1: nest, strict descent}} & \longrightarrow & \underbrace{\mathcal{A}_{\text{rel}}}_{\text{Stage 1.5: algebra-sheaf, flat}} \\ \xrightarrow{\text{Op}_{\text{NCG}}, \Omega_u} \underbrace{\mathcal{O}p_{\text{rel}}, \quad a_{ij} = dV_{ij}, \quad F = P(dP)^2}_{\text{Stage 2: represent+trace } T, \text{ universal } d} & \xrightarrow{\chi_\tau} & c_1^{\text{rel}} \in \check{H}^2(N(R_Q); \mathbb{Z}). \end{array}$$

1 Stage 1: the relational operator substrate

The substrate records what the local quantum world is made of, using observable algebras rather than a pre-selected sector, and it carries its relational comparisons as primitive data. It is built in two strictly flat layers: the relational operator nest that assigns an algebra to each patch and overlap (§1.1), and the descent of these into a sheaf of observable algebras over a comparison site (§1.2). Both layers are curvature-free: the comparisons compose on the nose, so the substrate is flat before it is spectral.

1.1 The relational operator nest

Let $\{U_i\}_{i \in I}$ be a relational cover with overlaps U_{ij} and triple overlaps U_{ijk} . The substrate is the *relational operator nest*

$$\mathcal{N}_{\text{rel}} = (\{A_i\}_{i \in I}, \{A_{ij}\}_{i,j}, \rho_{ij}^i, \rho_{ij}^j, \alpha_{ij}), \quad (1)$$

where

- A_i is the local observable algebra over the patch U_i , a unital C^* -algebra;
- A_{ij} is the observable algebra on the relational overlap U_{ij} ;
- $\rho_{ij}^i: A_i \rightarrow A_{ij}$ and $\rho_{ij}^j: A_j \rightarrow A_{ij}$ are the relational restriction/comparison maps;
- $\alpha_{ij}: A_i|_{ij} \xrightarrow{\sim} A_j|_{ij}$ is the comparison $*$ -isomorphism on the overlap;
- on triple overlaps the comparisons satisfy the strict cocycle condition

$$\alpha_{ij}\alpha_{jk} = \alpha_{ik} \quad \text{on } U_{ijk}. \quad (2)$$

The restriction maps ρ_{ij}^i and the comparisons α_{ij} are part of the data of the nest. We keep only the algebraic carrier role of a region — patches and overlaps carry observable algebras together with structured comparisons — without imposing microcausality, covariance, or a vacuum.

Remark 1.1 (Sector selection is downstream). The fibre A_i is the full observable algebra, not an occupied subspace. The Chern-sector reduction is downstream: after a realization one chooses a local Hamiltonian $H_i \in A_i$, forms the Fermi projection $P_i = \chi_{(-\infty, E_F)}(H_i)$, and the occupied subspace $E_i = \text{Ran } P_i$. Schematically,

$$\text{local observable algebra} \rightsquigarrow \text{chosen projection} \rightsquigarrow \text{Chern/NCG curvature}.$$

1.2 Stage 1.5: the algebra-sheaf by descent

The comparison data assemble directly into a sheaf. Over the comparison site $K = N(R_Q)$, the nerve of the relational groupoid $R_Q \rightrightarrows I$, and for a combinatorial region $V \subseteq K$, the relational algebra-sheaf assigns

$$\mathcal{A}_{\text{rel}}(V) = \{(a_i) : a_i \in A_i, \alpha_{ij}(a_i|_{ij}) = a_j|_{ij} \text{ on overlaps}\}. \quad (3)$$

This is the object of compatible local algebraic data; the assignment $V \mapsto \mathcal{A}_{\text{rel}}(V)$ is the relational sheaf,

$$\mathcal{A}_{\text{rel}}: U_i \longmapsto A_i, \quad \text{glued by the } \alpha_{ij}. \quad (4)$$

Because the restriction maps ρ_{ij}^i and the comparisons α_{ij} belong to the nest, the descent is immediate: the data glue on the nose and $\mathcal{A}_{\text{rel}}$ is flat — the comparisons compose strictly, so $\mathcal{A}_{\text{rel}}$ carries no holonomy. Whatever topology the system has is, at this stage, the combinatorial topology of the cover.

Remark 1.2 (The groupoid is a presentation). The comparison site is the nerve $K = N(R_Q)$, but R_Q enters only as bookkeeping for the cover and its overlaps. The descent data (A_i, α_{ij}) are foundational; the groupoid presents them.

Remark 1.3 (The defect is already in the descent site). At a $+\pi/3$ disclination the pentagon's isotropy carries five composable comparisons, not six, so the cover around the defect — and hence $K = N(R_Q)$ — is structurally altered there before any operator is assigned. The relational topology already sees the defect; the realization will only measure it.

2 Stage 2: the operator nest, the universal calculus, and the characteristic map

Only after the algebra-sheaf exists does one realize it as operators. This is the single place non-commutative geometry enters, and it enters lightly: as a representation that lets the algebra be differentiated and integrated, not as the injection of a metric operator.

The operator nest. The realization completes the algebra-sheaf into an operator sheaf,

$$\mathcal{O}_{\text{rel}} = \text{Op}_{\text{NCG}}(\mathcal{A}_{\text{rel}}), \quad \mathcal{O}_{\text{rel}}(V) = \text{End}_{\mathcal{A}_{\text{rel}}(V)}(\mathcal{E}(V)), \quad (5)$$

where $\mathcal{E}$ is the module on which the operators act. Concretely it represents the local algebras and implements the comparisons,

$$A_i \mapsto (\pi_i: A_i \rightarrow B(\mathcal{H}_i)), \quad \alpha_{ij} \mapsto V_{ij}, \quad V_{ij} \pi_i(a) V_{ij}^* = \pi_j(\alpha_{ij}(a)), \quad (6)$$

on $\mathcal{H} = \bigoplus_i \mathcal{H}_i$, with a faithful trace T . The occupied sector is selected inside the represented algebra, $P_i \in \pi_i(A_i)''$, $E_i = \text{Ran } P_i$. The realized sheaf is the compressed local calculus $U_i \mapsto P_i \Omega_u^\bullet P_i$; restriction is compression by sub-projectors, and gluing is by the implementers.

Definition 2.1 (Functorial implementer). V_{ij} is the operator that implements the comparison α_{ij} on the represented algebra,

$$V_{ij} \pi_i(a) V_{ij}^* = \pi_j(\alpha_{ij}(a)) \quad (a \in A_i),$$

determined by α_{ij} up to the unitary group of the commutant $\pi_j(A_j)'$. It is a representation-theoretic datum attached to the algebraic comparison α_{ij} ; its definition uses no Hamiltonian and no Fermi projection.

Proposition 2.2 (Implementers are not Berry/Kato transports). Let (V_{ij}) be functorial implementers of the strict cocycle (α_{ij}) , and — after a sector P_i has been chosen — let T_{ij} denote the Berry/Kato adiabatic parallel transport of the occupied subbundle $E_i = \text{Ran } P_i$. Then

- (i) the implementer cocycle is flat: $V_{ij} V_{jk} = V_{ik}$ strictly, hence the triangular holonomy $V_{ki} V_{jk} V_{ij} = 1$;
- (ii) the physical transport is, in general, not flat: its triangular holonomy $T_{ki} T_{jk} T_{ij} \neq 1$, the obstruction being the first Chern class of the occupied bundle;
- (iii) consequently $V_{ij} \neq T_{ij}$ in general. The Wilson/Kato links of the occupied bundle appear only downstream of sector selection, as the determinant phases on E_i ,

$$g_{ij} = \frac{\det(V_{ij}|_{E_i})}{|\det(V_{ij}|_{E_i})|},$$

whose holonomy is the realized curvature/defect class, not a datum of the descent cocycle.

Proof. (i) is functoriality of the realization applied to $\alpha_{ij} \alpha_{jk} = \alpha_{ik}$ and $\alpha_{ii} = \text{id}$. (ii) is the standard Berry/Wilson statement: adiabatic transport of a rank- r bundle over a triangulated base composes up to a $U(r)$ holonomy whose normalized trace over a 2-cycle integrates the Berry curvature, the integral obstruction being c_1 ; flatness would force $c_1 = 0$. (iii) follows: were $V_{ij} = T_{ij}$, parts (i) and (ii) would contradict whenever $c_1 \neq 0$. The determinant phase g_{ij} is the compression of the implementer to the occupied subspace, manifestly post- P_i , and is the discrete Wilson/Berry link whose product is read out downstream. $\square$

The differential is universal; the geometry is in the characteristic map. The differential of the calculus is not chosen by the realization. It is the universal differential calculus $\Omega_u^\bullet(A_G)$, $da = 1 \otimes a - a \otimes 1$ extended by the graded Leibniz rule: canonical, purely algebraic, and functorial. Functoriality is the point — every comparison α_{ij} lifts canonically to $\Omega_u(\alpha_{ij})$ intertwining d , and the strict cocycle lifts strictly, so the universal calculus is carried by the substrate, glued for free, and stays flat. It is also featureless: it injects no metric and no curvature of its own. Of everything the realization supplies, the one piece of analytic content needed downstream is the faithful trace T on A_G , which makes the integration converge; the invariant-selecting geometry — the position derivations $\partial_j = [X_j, \cdot]$ — lives not in the differential but in the cyclic characteristic map τ of §4. Hence the thesis of the paper: the noncommutative content of the theory is split between a free, universal differential and a chosen characteristic map; supply the position cocycle that defines τ , fix the trace, and the rest of the machinery turns automatically.

Remark 2.3 (Functoriality keeps the substrate flat). Because π_{NCG} preserves composition, $V_{ij}V_{jk} = V_{ik}$ holds strictly, so the triangular holonomy $V_{ki}V_{jk}V_{ij} = 1$ is trivial: a strict realization leaves no holonomy-curvature, and the universal calculus glues flatly. The universal curvature $F = P(dP)^2$ is therefore featureless until paired with τ ; all invariant content comes from the characteristic map. This is the point, not a defect: it commits the construction to the Chern–Weil picture and, as §3 shows, it is exactly what protects the free Bianchi identity. On a defect, the position cocycle that defines τ cannot use global coordinates — see §8.

3 Connection, curvature, and the automatic Bianchi identity

Curvature is born here, when d first acts on the selected projection. The connection is the universal derivative of the implementer and the curvature is its compressed square:

$$a_{ij} := dV_{ij}, \quad F = P(dP)^2, \quad F_{ijk}^{\text{rel}} = P_i a_{ij} a_{jk}. \quad (7)$$

Explicitly $F_{ijk}^{\text{rel}} = P_i dV_{ij} dV_{jk}$.

Proposition 3.1 (Bianchi is automatic). *The curvature (7) is closed without any associativity hypothesis.*

(i) *From $P^2 = P$ one gets $P(dP)P = 0$, and with Leibniz this is the Bianchi identity $\nabla F = 0$ for the Grassmann connection $\nabla = P d$.*

(ii) *From the strictly preserved composition $V_{ij}V_{jk} = V_{ik}$ and Leibniz,*

$$dV_{ik} = dV_{ij} V_{jk} + V_{ij} dV_{jk},$$

the Maurer–Cartan relation for a_{ij} , whence $\tilde{\delta} F_{ijk}^{\text{rel}} = 0$.

Proof. Differentiating $P^2 = P$ gives $(dP)P + P(dP) = dP$; compressing by P yields $P(dP)P = 0$, and the standard projector computation gives $\nabla F = 0$. Part (ii) is Leibniz applied to the functorial composition law of the implementers. $\square$

Closure is a calculus identity, and Proposition 3.1(ii) makes explicit that it is functoriality of π_{NCG} — preservation of $V_{ij}V_{jk} = V_{ik}$ — that carries it. No twisted algebra and no associativity theorem are needed.

4 The characteristic map and the relational Chern class

Closure of the universal curvature does not by itself yield a scalar; the bridge is a cyclic cocycle. This is the single, proper role of the characteristic map, and it is where the invariant is both selected and integrated.

Definition 4.1 (Characteristic map). *For a cyclic cocycle τ on A_G ($b\tau = 0$),*

$$\chi_\tau : F_{ijk}^{\text{rel}} \longmapsto c_{ijk} := \tau(F_{ijk}^{\text{rel}}) = \tau(P_i dV_{ij} dV_{jk}) \in \mathbb{C}. \quad (8)$$

For the Chern/Hall family τ is the Connes–Chern (Bellissard area) cocycle, of density $\frac{1}{2\pi i} \tau(P[\partial_1 P, \partial_2 P])$, $\partial_j = [X_j, \cdot]$. The position derivations ∂_j reside here, in τ — not in the differential — so the characteristic map is where the invariant-selecting geometry of the realization is carried.

Theorem 4.2 (Čech–Connes class and its integral lift). *If τ is a cyclic cocycle and F^{rel} is Bianchi-closed (Proposition 3.1), then $c = \chi_\tau(F^{\text{rel}})$ is a Čech 2-cocycle and defines $[c] \in \check{H}^2(N(R_Q); \mathbb{C})$ in the total cohomology of the Čech–Connes bicomplex (Appendix A). The complex class lifts to the integers in two complementary ways:*

- (i) *(bulk, absolute) the Connes pairing $\langle \tau, [P] \rangle \in \mathbb{Z}$ of the area cocycle with $[P] \in K_0(A_G)$ is an index, integral because τ is the Connes–Chern character of a finitely-summable Fredholm module;*
- (ii) *(defect, relative) for the rank-one occupied band the determinant phases g_{ij} are the $U(1)$ transition functions of the occupied line bundle, and the exponential sequence $U(1) \rightarrow \mathbb{R} \rightarrow \mathbb{Z}$ Bocksteins them on the disk $\text{St}(d)$, where $\check{H}^2(\text{St}(d), \partial \text{St}(d)) \cong \mathbb{Z}$.*

In both forms one obtains the relational first Chern class $c_1^{\text{rel}} \in \check{H}^2(N(R_Q); \mathbb{Z})$.

The integral lift is the point: a Chern class is quantized, and that is visible only over $\mathbb{Z}$. The bulk number is an index (needs the trace and the summable Fredholm structure); the defect number is a determinant winding (needs only the implementer determinants).

Remark 4.3 (The characteristic map is modular). The substrate, the realization, and Bianchi are fixed; the cyclic cocycle τ alone selects the invariant — area cocycle $\Rightarrow$ Chern/Hall, higher cocycles $\Rightarrow$ higher Chern, odd cocycles with a symmetry $\Rightarrow \mathbb{Z}_2$, the canonical trace $\Rightarrow$ gap-labeling, a quasi-energy $\tau \Rightarrow$ Floquet. An index cocycle gives a quantized integer; a non-index spectral functional gives non-quantized data (transport coefficients, resonances) on the same substrate. The differential and curvature never change; only χ_τ does.

5 The crystalline equivalence (foundational hypothesis)

The validation against known topology — that in the translation-invariant sector the relational class is the Bloch–Berry Chern class — remains unproved and is recorded as the foundational hypothesis.

Hypothesis 5.1 (Crystalline equivalence). *For the periodic Haldane model there is a translation-invariant sector $\check{H}^2(N(R_Q); \mathbb{Z})_{\text{cryst}} \subseteq \check{H}^2(N(R_Q); \mathbb{Z})$ and a comparison $\Theta : \check{H}^2(N(R_Q); \mathbb{Z})_{\text{cryst}} \rightarrow H^2(\mathbb{T}^2; \mathbb{Z})$ with $\Theta(c_1^{\text{rel}}) = c_1(B)$, $B \rightarrow \mathbb{T}^2$ the Bloch bundle, carrying the Connes–Chern density to the Berry curvature.*

Corollary 5.2. *Assume Hypothesis 5.1. Then $\langle \Theta(c_1^{\text{rel}}, [\mathbb{T}^2]) \rangle = \langle c_1(B), [\mathbb{T}^2] \rangle = C$, the Haldane Chern number.*

This is the external bridge to Bloch theory; it requires periodicity (a torus). The disclination class lies outside the crystalline sector and is reached without it.

6 The disclination and the relative class

Let d be a pentagonal disclination, $\text{St}(d) \subset N(R_Q)$ its star, $\partial \text{St}(d)$ its boundary. With c^{ref} the cochain of the defect-free reference, the relative cocycle $\rho_d = c - c^{\text{ref}}$ is supported on $\text{St}(d)$ when defect and reference agree outside it, and

$$c_d^{\text{rel}} := [\rho_d] \in \check{H}^2(\text{St}(d), \partial \text{St}(d)), \quad \Phi_d = \prod_{(ij) \in L_d} g_{ij}, \quad g_{ij} = \frac{\det(V_{ij}|_{E_i})}{|\det(V_{ij}|_{E_i})|}, \quad (9)$$

the pairing of c_d^{rel} with the loop L_d around the pentagon (Figure 1). Here, by Proposition 2.2(iii), the g_{ij} are the Wilson/Berry determinant phases of the implementers compressed to the occupied subbundle E_i , hence post-sector-selection readouts rather than Stage-1 data.

The next proposition isolates why a flat substrate can nonetheless carry a nonzero Φ_d : strict composability is a bulk statement, and the disclination is a controlled, localized violation of it.

Proposition 6.1 (Bulk strictness vs. defect holonomy). *Suppose the strict cocycle $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$ holds on every bulk triple overlap, so that $V_{ij}V_{jk} = V_{ik}$ there. Then:*

- (i) *around any bulk loop the implementers telescope and $\Phi = 1$ identically;*
- (ii) *a nonzero pentagon holonomy $\Phi_d \neq \Phi_d^{\text{ref}}$ requires the loop L_d to fail strict closure — which is exactly the structural alteration of the descent site at d (five composable comparisons, not six), the disclination conjugating the incident links by the Frank-wedge rotation $R(\pi/3)$;*
- (iii) *hence Φ_d measures the localized failure of strict closure at d ; it is a transport-determinant (post- P_i , independent of τ) quantity, consistent with the substrate being flat in the bulk.*

Proof. (i) Multiplicativity of $\det$ on E_i and strict composition give $\prod_{(ij) \in L} g_{ij} = \det(\prod_L V_{ij}|_E)/|\cdot|$; a closed bulk loop composes to the identity, so the determinant phase is 1. (ii) If $V_{ij}V_{jk} = V_{ik}$ held around L_d as well, the same telescoping would force $\Phi_d = 1 = \Phi_d^{\text{ref}}$; a nonzero relative phase therefore obstructs strict closure on L_d , which is the five-fold coordination of Remark 1.4 and the $R(\pi/3)$ conjugation. (iii) is the combination, with the characteristic map's derivations entering nowhere in g_{ij} . $\square$

7 The Local Defect Theorem

Theorem 7.1 (Local Defect Theorem). *Assume bulk flatness: $F = F^{\text{ref}}$ and $P = P^{\text{ref}}$ off $\text{St}(d)$. Then c_d^{rel} is gauge- and Morita-invariant, supported on $\text{St}(d)$, and*

$$[c_d] \neq 0 \iff c_d^{\text{rel}} \neq 0 \iff \Phi_d \neq \Phi_d^{\text{ref}}.$$

Moreover the deciding quantity Φ_d is built from transport determinants g_{ij} alone, so for the determinant (Chern) characteristic map the defect decision is independent of the characteristic map's derivations.

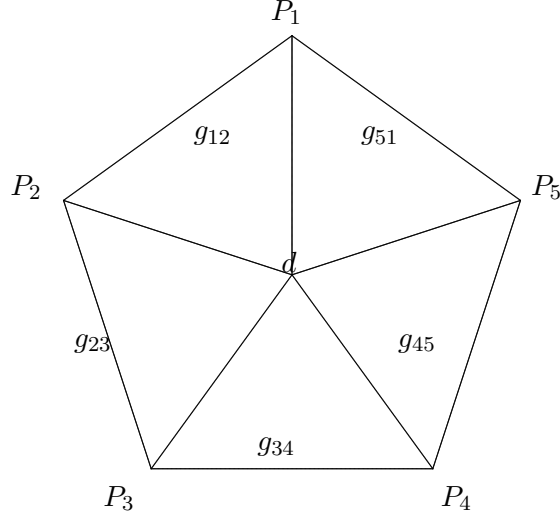


Figure 1: The star $\text{St}(d)$: five contexts, five implementers, five triangles $\{d, i, i+1\}$ carrying $F_{d,i,i+1}^{\text{rel}}$. Curvature is supported here; c_1^{rel} localizes as c_d^{rel} , paired with L_d as $\Phi_d = \prod_i g_{i,i+1}$.

Sketch. Invariance: $[c]$ is a Čech–Connes class; a frame change shifts a_{ij} by an inner term traced into a coboundary by cyclicity of τ , and a realization change presenting the same R_Q is Morita. Localization: bulk flatness makes ρ_d vanish off $\text{St}(d)$; the determinant phases telescope along interior transports, leaving the boundary ratio $\Phi_d/\Phi_d^{\text{ref}}$. The last clause follows because $g_{ij} = \det(V_{ij}|_{E_i})/|\det(V_{ij}|_{E_i})|$ does not involve the position derivations. $\square$

So the defect monodromy is a transport-level quantity (post-functor, post- P_i , independent of τ), while the bulk Chern class is an index/characteristic-map-level quantity; this theorem is the bridge that lets the former decide the latter.

8 Two open constructions

The two-stage formulation concentrates all genuine difficulty in two places.

Open construction 8.1 (The position cocycle and the characteristic map). *On a disclination there is no single-valued position, so the derivations cannot be the flat-plane ∂_X, ∂_Y . They must be built from a position 1-cocycle on the groupoid, $X : R_Q \rightarrow \mathbb{R}^2$ with $X(\gamma\gamma') = X(\gamma) + \gamma \cdot X(\gamma')$, giving relational derivations $\partial_j = [X_j, \cdot]$ without global coordinates. These derivations, together with the trace T , define the area cocycle τ — the characteristic map — and so, by the thesis of §2, carry the entire invariant-selecting geometry. What the construction pivots on is the existence of the position cocycle and the finiteness/summability of T that makes $\langle \tau, [P] \rangle$ land in $\mathbb{Z}$; no self-adjoint Dirac operator and no global coordinates are required.*

Open construction 8.2 (The realization comparison). π_{NCG} induces a map on cohomology; one must show it is an isomorphism on the sector of interest, so that the relational Čech class and the realized NCG class agree. Equivalently, the bulk Bellissard curvature $P[\partial_1 P, \partial_2 P]$ and the simplicial

F_{ijk}^{rel} represent the same Čech–Connes class. This comparison is the one nontrivial internal identity; it is periodicity-free, and distinct from Hypothesis 5.1 (which is the external bridge to Bloch theory).

9 To-do list: computing the Haldane disclination

For the Haldane model in the topological regime ($C = \pm 1$) the pipeline is finite.

Stage 1 — local algebra substrate.

- Attach local observable algebras A_i around the defect. Record the comparison $*$ -isomorphisms α_{ij} on overlaps and the descent site $K = N(R_Q)$, including the five-fold star at d . No sector is selected at this stage: H_i, P_i, E_i are downstream (§2).

Stage 2 — realize, select sector, and differentiate.

- Apply π_{NCG} : $\pi_i : A_i \rightarrow B(\mathcal{H}_i)$, $\alpha_{ij} \mapsto V_{ij}$ (functorial implementers from the descent isomorphisms, not Berry/Kato transports; Definition 2.1, Proposition 2.2), $R_Q \mapsto A_G$ with faithful trace T .
- After realization/readout, select $H_i \in A_i$, form the Fermi projection $P_i = \chi_{(-\infty, E_F)}(H_i) \in \pi_i(A_i)''$, and set $E_i = \text{Ran } P_i$ (frames e_i if used).
- Take the universal differential d (free, functorial); build the position cocycle X and derivations $\partial_j = [X_j, \cdot]$ defining τ (Open construction 8.1); form $a_{ij} = dV_{ij}$ and $F_{ijk}^{\text{rel}} = P_i a_{ij} a_{jk}$.
- Closure is structural: verify $dV_{ik} = dV_{ij} V_{jk} + V_{ij} dV_{jk}$ (Proposition 3.1); no associativity check.

Stage 3 — characteristic map and class.

- Apply the Connes–Chern area cocycle: $c_{ijk} = \tau(F_{ijk}^{\text{rel}})$; integralize the bulk number as the index pairing $\langle \tau, [P] \rangle$ to get $c_1^{\text{rel}} = [c] \in \check{H}^2(N(R_Q); \mathbb{Z})$.
- Cross-check $\langle c_1^{\text{rel}}, [\Sigma] \rangle$ against the bulk Bellissard number $\frac{1}{2\pi i} \tau(P[\partial_1 P, \partial_2 P])$.

Stage 4 — localize and decide (transport-level).

- On $\text{St}(d)$ form $\rho_d = c - c^{\text{ref}}$ (nonzero by five- vs. six-fold coordination; Proposition 6.1); $\text{St}(d)$ is a disk so $\check{H}^2(\text{St}(d), \partial \text{St}(d)) \cong \mathbb{Z}$.
- Compute the links $g_{i,i+1} = \det(V_{i,i+1}|_{E_i}) / |\det(V_{i,i+1}|_{E_i})|$ and $\Phi_d = \prod_i g_{i,i+1}$, Φ_d^{ref} . The disclination conjugates the incident $V_{i,i+1}$ by $R(\pi/3)$ (the Frank wedge).
- Decide: $\Delta\Phi = \arg(\Phi_d / \Phi_d^{\text{ref}}) \in \mathbb{R}/2\pi\mathbb{Z}$, with $c_d^{\text{rel}} \neq 0 \iff \Delta\Phi \notin 2\pi\mathbb{Z}$. By Theorem 7.1 this step needs only the transports, not the characteristic map's derivations; the expected form is $\Delta\Phi \propto (\text{Frank angle}/2\pi)$ weighted by bulk band data.

10 Status and outlook

Relations come first and are flat; the realization functor represents the algebra and supplies a trace, the differential is the universal calculus carried for free by the substrate, and curvature is

born only when d acts on the selected projection. The geometry that distinguishes invariants enters at one point — the cyclic characteristic map, whose position derivations live in τ , not in any Dirac operator. Closure is then a calculus identity that functoriality protects (Proposition 3.1), and a single cyclic cocycle bridges the universal curvature to a Čech–Connes class and selects the invariant. The crystalline equivalence (Hypothesis 5.1) remains the open external validation against Bloch theory and is not assumed. All genuine difficulty is concentrated in the position cocycle defining the characteristic map (Open construction 8.1) and the realization comparison (Open construction 8.2); the disclination decision, by contrast, reduces to a finite transport-determinant holonomy.

A The Čech–Connes bicomplex

The cover direction supplies the Čech differential $\check{\delta}$ on $N(R_Q)$; the operator direction supplies the cyclic (b, B) differentials of $\Omega_u^\bullet(A_G)$. Curvature is a total representative: F^{rel} is $\check{\delta}$ -closed by Bianchi (Proposition 3.1) and τ is b -closed; χ_τ contracts the operator legs, leaving a $\check{\delta}$ -cocycle of class $[c] \in \check{H}^2(N(R_Q); \mathbb{C})$, integralized to c_1^{rel} by the index pairing (bulk) or the exponential sequence (defect). This is the standard setting for a relational NCG Chern–Weil statement: noncommutative curvature, abelian cohomology, mediated by a cyclic characteristic map [1,2,5].

B Dictionary: relational $\rightarrow$ realized $\rightarrow$ Haldane

Relational (Stage 1 / 1.5)	NCG realization (Stage 2)	Haldane
local algebra A_i	represented $\pi_i(A_i) \subset B(\mathcal{H}_i)$	local observable algebra
selected sector $P_i \in \pi_i(A_i)''$	$E_i = \text{Ran } P_i$	Fermi projection $\chi_{(-\infty, E_F)}(H_i)$
comparison α_{ij} (strict $*$ -iso)	implementer $V_{ij}, V_{ij}V_{jk} = V_{ik}$	strict implementer (not Berry/Kato transport)
descent site $K = N(R_Q)$	algebra A_G , trace T , universal Ω_u	observables, position cocycle
sheaf $\mathcal{A}_{\text{rel}}(U_i) = A_i$ (per-patch, glued by α_{ij})	$P_i \Omega_u^\bullet(A_G) P_i$	local calculi
(flat: strict descent)	connection $a_{ij} = dV_{ij}$	transport derivative
—	curvature $F = P(dP)^2$	$\pi/3$ at the pentagon
combinatorial closure	$P^2 = P$ & Leibniz $\Rightarrow \check{\delta}F = 0$	automatic Bianchi
—	χ_τ (Connes–Chern), $\partial_j = [X_j, \cdot]$	area cocycle
relational class $[c] \in \check{H}^2$; lift $c_1^{\text{rel}} \in \check{H}^2(\mathbb{Z})$	$\langle \tau, [P] \rangle$ index pairing; defect c_d^{rel}	$\Phi_d = \prod g_{ij}$, Wilson dets on E_i (post- P_i)

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